ADITYA PATIYAL

Full-Stack Developer · **Computer Vision Engineer** · **AI / ML Practitioner**
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SUMMARY

3rd-year CS undergrad (AI & ML) at Manipal University Jaipur specialising in full-stack development and computer vision. Ranked **top 500 globally** in Amazon ML Hackathon and **top 5** at university IEEE Hackathon. Solved **800+ DSA problems** on LeetCode/CodeChef; Codeforces rating 1300.

EDUCATION

Manipal University Jaipur 2023 – 2027
B.Tech in Computer Science Engineering (AI & ML)

ACHIEVEMENTS

- Ranked **top 500 globally** in the Amazon ML Global Hackathon.
- Secured **top 5** at Manipal University Jaipur's IEEE Hackathon.
- Selected for the **HSBC Business Innovation Program** — a competitive multi-round hackathon bridging technology and business strategy.

CERTIFICATIONS

Machine Learning Specialization — Stanford University & DeepLearning.AI (Coursera) 2026
Deep Learning & Machine Learning — NPTEL (IIT-backed, Gold Certified) 2025

PROJECTS

Legal AI Platform | GitHub | Demo | FastAPI · LangChain · ChromaDB · Groq Llama-3.3-70B · Docker · Railway

- An AI-powered platform where users can **upload any legal document**, ask questions in plain English, and instantly get clause-by-clause explanations, risk scores, and ready-to-download legal drafts — no lawyer needed for initial review.
- Engineered hybrid RAG retrieval (ChromaDB semantic + BM25 lexical) with a dual-LLM pipeline (Groq cloud + Ollama offline); containerised with Docker and deployed on Railway + Netlify.

Multimodal Desktop Controller | GitHub | Demo | Python · SpeechRecognition · Tkinter · pynput · AppleScript

- A **hands-free desktop assistant** that lets you control your computer entirely through voice and gestures — switch modes by activity (coding, media, or deep focus), and the system automatically decides whether to listen to your voice or watch your hands.
- Built a modular 3-mode state machine (CODING / MEDIA / FOCUS) with NLP intent parsing and a real-time Tkinter overlay for live system-state feedback. (In progress)

Jaipur Tourism Platform | GitHub | Demo | Node.js · Express.js · Gemini API · JWT · Bcryptjs · Vercel

- A full travel companion for Jaipur visitors — users can **discover places, filter hotels and events**, get a personalised day-by-day itinerary from an AI planner, chat with a travel assistant, and even kill time with built-in games, all from one site.
- 50+ destination pages with Gemini-powered AI chat assistant, JWT-authenticated user accounts for saved itineraries, and advanced filtering by area, rating, and category; deployed on Vercel.

AirPad — Gesture Virtual Touchpad | GitHub | Demo | Python · OpenCV · MediaPipe · PyAutoGUI

- Turns your laptop camera into a **touchless mouse** — point a finger to move the cursor, pinch to click, use two fingers to scroll, and tilt your left hand to zoom, all without touching the trackpad.
- Tracks 21 hand landmarks in real time via MediaPipe; implements smooth cursor acceleration, inertia scrolling, dwell-click (3s hover = click with progress arc), and a live overlay showing current gesture mode.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, HTML5, CSS3
Frameworks & Web: React.js, Node.js, Express.js, FastAPI, Tailwind CSS
AI & ML: LangChain, OpenCV, MediaPipe, NumPy, Pandas, scikit-learn, RAG, LLM APIs (Groq, Ollama, Gemini)
Databases: MongoDB, ChromaDB, SQLite, Firebase
DevOps & Tools: Git, GitHub, Docker, Postman, VS Code, WebRTC, Socket.io, Vercel, Railway